

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

WEEK - 3

Task - 1

Step -1: Open Terminal On Mac

Step -2: Check Whether MongoDB Shell is installed

Check - mongosh

Connect the server:

```
Last login: Fri Aug 14 09:19:39 on console
[vsameerreddy@Sameer-Reddy ~ % mongosh "mongodb+srv://cluster0.erhexq0.mongodb.net/" --apiVersion 1 --username vemireddys]
ameerreddy_db_user
[Enter password: *****]
Current Mongosh Log ID: 6a83c923f591f08bfc667342
Connecting to:      mongodb+srv://<credentials>@cluster0.erhexq0.mongodb.net/?appName=mongosh+2.9.2
Using MongoDB:      8.0.29 (API Version 1)
Using Mongosh:      2.9.2

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/
```

Step -3: Create And Use the database

```
[Atlas atlas-dklk63-shard-0 [primary] test> use bookflow_db
switched to db bookflow_db
[Atlas atlas-dklk63-shard-0 [primary] bookflow_db> db
bookflow_db
```

Step -4: Create the first book document

```
Atlas atlas-dklk63-shard-0 [primary] bookflow_db> db.book_metadata.insertOne({
  |   title: "The Silent Algorithm",
  |   author: "R. Menon",
  |   genre: "Technology",
  |   publishedYear: 2022,
  |   formats: ["Hardcover", "Digital", "Audiobook"],
  |   specs: {
  |     pages: 340,
  |     fileSize_MB: 4.2,
  |     drm: true,
  |     narrator: "K. Rao",
  |     runtime_min: 610
  |   },
  |   avgRating: 4.6,
  |   reviews: [
  |     {
  |       member_id: "M1001",
  |       rating: 5,
  |       comment: "Excellent deep dive into digital systems design.",
  |       date: ISODate("2023-01-15")
  |     },
  |     {
  |       member_id: "M1042",
  |       rating: 4,
  |       comment: "Good but the Digital edition had formatting issues.",
  |       date: ISODate("2023-02-20")
  |     }
  |   ]
  | })
{}
```

```
{
  acknowledged: true,
  insertedId: ObjectId('6a83ccb7f591f08bfc667343')
}
```

Step -5:Insert the remaining books

```
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
    }
}
]
}
]
Atlas atlas-dklk63-shard-0 [primary] bookflow_db> db.book_metadata.insertMany([
| {
|   title: "Ocean of Code",
|   author: "A. Fernandes",
|   genre: "Technology",
|   publishedYear: 2021,
|   formats: ["Digital", "Paperback"],
|   specs: {
|     pages: 210,
|     fileSize_MB: 2.1,
|     drm: false
|   },
|   avgRating: 4.2,
|   reviews: [
|     {
|       member_id: "M2001",
|       rating: 4,
|       comment: "Solid Digital read, clean layout.",
|       date: ISODate("2022-05-10")
|     },
|     {
|       member_id: "M2002",
|       rating: 5,
|
|       rating: 5,
|       comment: "Best programming book this year.",
|       date: ISODate("2022-06-01")
|     }
|   ],
| },
| {
|   title: "Whispers of Yesterday",
|   author: "S. Kapoor",
|   genre: "Fiction",
|   publishedYear: 2019,
|   formats: ["Hardcover", "Paperback"],
|   specs: {
|     pages: 410,
|     binding: "Sewn"
|   },
|   avgRating: 3.8,
|   reviews: [
|     {
|       member_id: "M3001",
|       rating: 3,
|       comment: "Slow start but strong ending.",
|       date: ISODate("2020-01-15")
|     },
|     {
|       member_id: "M3002",
|       rating: 4,
|       comment: "Beautifully written prose.",
|       date: ISODate("2020-02-11")
|     }
|   ],
| },
| {
|   title: "Quantum Horizons",
|   author: "R. Menon",
|   genre: "Science",
|   publishedYear: 2023,
|   formats: ["Digital", "Audiobook"],
|   specs: {
|     fileSize_MB: 5.6,
|     drm: true,
|     narrator: "P. Iyer",
|     runtime_min: 540
|   },
|   avgRating: 4.9,
|   reviews: [
|     {
|       member_id: "M4001",
|       rating: 5,
|       comment: "The Digital format made annotations so easy.",
|       date: ISODate("2023-08-01")
|     },
|     {
|       member_id: "M4002",
|       rating: 5,
|       comment: "Mind-bending and well narrated.",
|       date: ISODate("2023-09-14")
|     }
|   ],
| },
| ]
| })
```

Step -6: Check all inserted data

```
[Atlas atlas-dklk63-shard-0 [primary] bookflow_db> db.book_metadata.find()
[
  {
    _id: ObjectId('6a83ccb7f591f08bfc667343'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a83cd06f591f08bfc667344'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      },
      {
        member_id: 'M2002',
        rating: 5,
        comment: 'Best programming book this year.',
        date: ISODate('2022-06-01T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a83cd06f591f08bfc667345'),
    title: 'Whispers of Yesterday',
    author: 'S. Kapoor',
    genre: 'Fiction',
    publishedYear: 2019,
    formats: [ 'Hardcover', 'Paperback' ],
    specs: { pages: 410, binding: 'Sewn' },
    avgRating: 3.8,
    reviews: [
      {
        member_id: 'M3001',
        rating: 3,
        comment: 'Slow start but strong ending.',
        date: ISODate('2020-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M3002',
        rating: 4,
        comment: 'Beautifully written prose.',
        date: ISODate('2020-02-11T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a83cd06f591f08bfc667346'),
    title: 'Quantum Horizons',
    author: 'R. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      },
      {
        member_id: 'M4002',
        rating: 5,
        comment: 'Mind-bending and well narrated.',
        date: ISODate('2023-09-14T00:00:00.000Z')
      }
    ]
  }
]
```

```

[Atlas atlas-dklk63-shard-0 [primary] bookflow_db> db.book_metadata.find().pretty()
[
  {
    _id: ObjectId('6a83ccb7f591f08bfc667343'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a83cd06f591f08bfc667344'),
    title: 'Ocean of Code',
    author: 'A. Fernandes',
    genre: 'Technology',
    publishedYear: 2021,
    formats: [ 'Digital', 'Paperback' ],
    specs: { pages: 210, fileSize_MB: 2.1, drm: false },
    avgRating: 4.2,
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      },
      {
        member_id: 'M2002',
        rating: 5,
        comment: 'Best programming book this year.',
        date: ISODate('2022-06-01T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a83cd06f591f08bfc667345'),
    title: 'Whispers of Yesterday',
    author: 'S. Kapoor',
    genre: 'Fiction',
    publishedYear: 2019,
    formats: [ 'Hardcover', 'Paperback' ],
    specs: { pages: 410, binding: 'Sewn' },
    avgRating: 3.8,
    reviews: [
      {
        member_id: 'M3001',
        rating: 3,
        comment: 'Slow start but strong ending.',
        date: ISODate('2020-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M3002',
        rating: 4,
        comment: 'Beautifully written prose.',
        date: ISODate('2020-02-11T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a83cd06f591f08bfc667346'),
    title: 'Quantum Horizons',
    author: 'R. Menon',
    genre: 'Science',
    publishedYear: 2023,
    formats: [ 'Digital', 'Audiobook' ],
    specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      },
      {
        member_id: 'M4002',
        rating: 5,
        comment: 'Mind-bending and well narrated.',
        date: ISODate('2023-09-14T00:00:00.000Z')
      }
    ]
  }
]

```

TASK 2: Advanced Querying

Step 7: Books rated above 4 -

```
Atlas atlas-dklk63-shard-0 [primary] bookflow_db> db.book_metadata.find(
|   { avgRating: { $gt: 4 } },
|   { title: 1, avgRating: 1, _id: 0 }
| )
[
|   { title: 'The Silent Algorithm', avgRating: 4.6 },
|   { title: 'Ocean of Code', avgRating: 4.2 },
|   { title: 'Quantum Horizons', avgRating: 4.9 }
| ]
```

Step 8: Find books in Digital or Audiobook format-

```
Atlas atlas-dklk63-shard-0 [primary] bookflow_db> db.book_metadata.find(
|   { formats: { $in: ["Digital", "Audiobook"] } },
|   { title: 1, formats: 1, _id: 0 }
| )
[
|   {
|     title: 'The Silent Algorithm',
|     formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
|   },
|   { title: 'Ocean of Code', formats: [ 'Digital', 'Paperback' ] },
|   { title: 'Quantum Horizons', formats: [ 'Digital', 'Audiobook' ] }
| ]
```

Step 9: Books with rating above 4 AND Digital format

```
Atlas atlas-dklk63-shard-0 [primary] bookflow_db> db.book_metadata.find({
|   avgRating: { $gt: 4 },
|   formats: { $in: ["Digital"] }
| })
[ ]
[
|   {
|     _id: ObjectId('6a83ccb7f591f08bfc667343'),
|     title: 'The Silent Algorithm',
|     author: 'R. Menon',
|     genre: 'Technology',
|     publishedYear: 2022,
|     formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
|     specs: {
|       pages: 340,
|       fileSize_MB: 4.2,
|       drm: true,
|       narrator: 'K. Rao',
|       runtime_min: 610
|     },
|     avgRating: 4.6,
|     reviews: [
|       {
|         member_id: 'M1001',
|         rating: 5,
|         comment: 'Excellent deep dive into digital systems design.',
|         date: ISODate('2023-01-15T00:00:00.000Z')
|       },
|       {
|         member_id: 'M1042',
|         rating: 4,
|         comment: 'Good but the Digital edition had formatting issues.',
|         date: ISODate('2023-02-20T00:00:00.000Z')
|       }
|     ]
|   },
|   {
|     _id: ObjectId('6a83cd06f591f08bfc667344'),
|     title: 'Ocean of Code',
|     author: 'A. Fernandes',
|     genre: 'Technology',
|     publishedYear: 2021,
|     formats: [ 'Digital', 'Paperback' ],
|     specs: { pages: 210, fileSize_MB: 2.1, drm: false },
|     avgRating: 4.2,
|     reviews: [
|       {
|         member_id: 'M2001',
|         rating: 4,
|         comment: 'Solid Digital read, clean layout.',
|         date: ISODate('2022-05-10T00:00:00.000Z')
|       },
|       {
|         member_id: 'M2002',
|         rating: 5,
|         comment: 'Best programming book this year.',
|         date: ISODate('2022-06-01T00:00:00.000Z')
|       }
|     ]
|   }
| ]
```

```

{
  _id: ObjectId('6a83cd06f591f08bfc667346'),
  title: 'Quantum Horizons',
  author: 'R. Menon',
  genre: 'Science',
  publishedYear: 2023,
  formats: [ 'Digital', 'Audiobook' ],
  specs: { fileSize_MB: 5.6, drm: true, narrator: 'P. Iyer', runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    },
    {
      member_id: 'M4002',
      rating: 5,
      comment: 'Mind-bending and well narrated.',
      date: ISODate('2023-09-14T00:00:00.000Z')
    }
  ]
}
]

```

Step 10: Find books with “Digital” Using \$regex and Find reviews mentioning “Digital”

```

Atlas atlas-dklk63-shard-0 [primary] bookflow_db> db.book_metadata.find(
| { formats: { $regex: "Digital", $options: "i" } },
| { title: 1, formats: 1 }
|)
[
  {
    _id: ObjectId('6a83ccb7f591f08bfc667343'),
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  },
  {
    _id: ObjectId('6a83cd06f591f08bfc667344'),
    title: 'Ocean of Code',
    formats: [ 'Digital', 'Paperback' ]
  },
  {
    _id: ObjectId('6a83cd06f591f08bfc667346'),
    title: 'Quantum Horizons',
    formats: [ 'Digital', 'Audiobook' ]
  }
]
Atlas atlas-dklk63-shard-0 [primary] bookflow_db> db.book_metadata.find(
| { "reviews.comment": { $regex: "Digital", $options: "i" } },
| { title: 1, "reviews.$": 1 }
|)
[
  {
    _id: ObjectId('6a83ccb7f591f08bfc667343'),
    title: 'The Silent Algorithm',
    reviews: [

```

```

      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a83cd06f591f08bfc667344'),
    title: 'Ocean of Code',
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: ISODate('2022-05-10T00:00:00.000Z')
      }
    ]
  },
  {
    _id: ObjectId('6a83cd06f591f08bfc667346'),
    title: 'Quantum Horizons',
    reviews: [
      {
        member_id: 'M4001',
        rating: 5,
        comment: 'The Digital format made annotations so easy.',
        date: ISODate('2023-08-01T00:00:00.000Z')
      }
    ]
  }
]

```

```

    }
  ]
},
{
  _id: ObjectId('6a83cd06f591f08bfc667346'),
  title: 'Quantum Horizons',
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: ISODate('2023-08-01T00:00:00.000Z')
    }
  ]
}
]
}
]

```

TASK 3

Step 11: Run the complete Aggregation Pipeline

```

Atlas atlas-dk1k63-shard-0 [primary] bookflow_db> db.book_metadata.aggregate([
| {
|   $match: {
|     publishedYear: { $gt: 2020 }
|   },
|   {
|     $unwind: "$reviews"
|   },
|   {
|     $group: {
|       _id: "$title",
|       avgReviewRating: { $avg: "$reviews.rating" },
|       totalReviews: { $sum: 1 },
|       author: { $first: "$author" }
|     },
|     {
|       $sort: {
|         avgReviewRating: -1
|       }
|     },
|     {
|       $project: {
|         _id: 0,
|         title: "$_id",
|         author: 1,
|         avgReviewRating: {
|           $round: ["$avgReviewRating", 2]
|         },
|       },
|       totalReviews: 1
|     }
|   }
| ])
| [
| {
|   totalReviews: 2,
|   author: 'R. Menon',
|   title: 'Quantum Horizons',
|   avgReviewRating: 5
| },
| {
|   totalReviews: 2,
|   author: 'R. Menon',
|   title: 'The Silent Algorithm',
|   avgReviewRating: 4.5
| },
| {
|   totalReviews: 2,
|   author: 'A. Fernandes',
|   title: 'Ocean of Code',
|   avgReviewRating: 4.5
| }
| ]

```

TASK 4: INDEXING FOR SPEED

Step 12: Create a text index on review comments

```
Atlas atlas-dklk63-shard-0 [primary] bookflow_db> db.book_metadata.createIndex({
|   "reviews.comment": "text"
| })
reviews.comment_text
```

Step 13: Search using the text index

```
Atlas atlas-dklk63-shard-0 [primary] bookflow_db> db.book_metadata.find(
| { $text: { $search: "Digital" } },
| {
|   title: 1,
|   score: { $meta: "textScore" }
| }
| ).sort({
|   score: { $meta: "textScore" }
| })
| )
|
| {
|   _id: ObjectId('6a83ccb7f591f08bfc667343'),
|   title: 'The Silent Algorithm',
|   score: 1.1833333333333333
| },
| {
|   _id: ObjectId('6a83cd06f591f08bfc667344'),
|   title: 'Ocean of Code',
|   score: 0.6
| },
| {
|   _id: ObjectId('6a83cd06f591f08bfc667346'),
|   title: 'Quantum Horizons',
|   score: 0.6
| }
| ]
```

Step 14: Check Whether the text index is being used

```
Atlas atlas-dklk63-shard-0 [primary] bookflow_db> db.book_metadata.find(
| { $text: { $search: "Digital" } }
| ).explain("executionStats")
| )
| {
|   explainVersion: '1',
|   queryPlanner: {
|     namespace: 'bookflow_db.book_metadata',
|     parsedQuery: {
|       '$text': {
|         '$search': 'Digital',
|         '$language': 'english',
|         '$caseSensitive': false,
|         '$diacriticSensitive': false
|       }
|     },
|     indexFilterSet: false,
|     queryHash: 'CF6F4CEE',
|     planCacheShapeHash: 'CF6F4CEE',
|     planCacheKey: '08852285',
|     optimizationTimeMillis: 0,
|     maxIndexedOrSolutionsReached: false,
|     maxIndexedAndSolutionsReached: false,
|     maxScansToExplodeReached: false,
|     prunedSimilarIndexes: false,
|     winningPlan: {
|       isCached: false,
|       stage: 'TEXT_MATCH',
|       indexPrefix: {},
|       indexName: 'reviews.comment_text',
|       parsedTextQuery: {
```

```
    terms: [ 'digit' ],
    negatedTerms: [],
    phrases: [],
    negatedPhrases: []
  },
  textIndexVersion: 3,
  inputStage: {
    stage: 'FETCH',
    inputStage: {
      stage: 'IXSCAN',
      keyPattern: { _fts: 'text', _ftsx: 1 },
      indexName: 'reviews.comment_text',
      isMultiKey: true,
      isUnique: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: 'backward',
      indexBounds: {}
    }
  },
  rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 0,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
    isCached: false,
    stage: 'TEXT_MATCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    indexPrefix: {},
    indexName: 'reviews.comment_text',
    parsedTextQuery: {
      terms: [ 'digit' ],
      negatedTerms: [],
      phrases: [],
      negatedPhrases: []
    },
    textIndexVersion: 3,
    docsRejected: 0,
    inputStage: {
      stage: 'FETCH',
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 4,
      advanced: 3,
      needTime: 0,
      needYield: 0,
```

```

saveState: 0,
restoreState: 0,
isEOF: 1,
docsExamined: 3,
alreadyHasObj: 0,
inputStage: {
  stage: 'IXSCAN',
  nReturned: 3,
  executionTimeMillisEstimate: 0,
  works: 4,
  advanced: 3,
  needTime: 0,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  keyPattern: { _fts: 'text', _ftsx: 1 },
  indexName: 'reviews.comment_text',
  isMultiKey: true,
  isUnique: false,
  isSparse: false,
  isPartial: false,
  indexVersion: 2,
  direction: 'backward',
  indexBounds: {},
  keysExamined: 3,
  seeks: 1,
  dupsTested: 3,
  dupsDropped: 0
}

```

```

queryShapeHash: '3EEB4213C3DA7110EC07CC52F25CC0ACA6ADE44C1D3B8434D5E58A19A41AF940',
command: {
  find: 'book_metadata',
  filter: { '$text': { '$search': 'Digital' } },
  '$db': 'bookflow_db'
},
serverInfo: {
  host: 'ac-5e4ljmv-shard-00-02.erhexq0.mongoddb.net',
  port: 27017,
  version: '8.0.29',
  gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1787022814, i: 3 }),
  signature: {
    hash: Binary.createFromBase64('K845JlweIQ6Uuaqvscg+Yqg3RHc=', 0),
    keyId: Long('7632047810002550786')
  }
}

```

```

},
operationTime: Timestamp({ t: 1787022814, i: 3 })
}
Atlas atlas-dklk63-shard-0 [primary] bookflow_db> db.book_metadata.find(
| {
|   "reviews.comment": {
|     $regex: "Digital",
|     $options: "i"
|   }
| }
| )
| )explain("executionStats")
| {
|   explainVersion: '1',
|   queryPlanner: {
|     namespace: 'bookflow_db.book_metadata',
|     parsedQuery: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
|     indexFilterSet: false,
|     queryHash: '9DF7CB16',
|     planCacheShapeHash: '9DF7CB16',
|     planCacheKey: 'A9183CB6',
|     optimizationTimeMillis: 0,
|     maxIndexedOrSolutionsReached: false,
|     maxIndexedAndSolutionsReached: false,
|     maxScansToExplodeReached: false,
|     prunedSimilarIndexes: false,
|     winningPlan: {
|       isCached: false,
|       stage: 'COLLSCAN',
|       filter: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },

```

Step 17: Create an index on formats

```
Atlas atlas-dklk63-shard-0 [primary] bookflow_db> db.book_metadata.createIndex({
  |   formats: 1
  | })
formats_1
```

Step 18: Create an index on avgRating

```
Atlas atlas-dklk63-shard-0 [primary] bookflow_db> db.book_metadata.createIndex({
  |   avgRating: -1
  | })
avgRating_-1
```

Step 19: Check the avgRating index

```
Atlas atlas-dklk63-shard-0 [primary] bookflow_db> db.book_metadata.find({
  |   avgRating: { $gt: 4 }
  | }).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflow_db.book_metadata',
    parsedQuery: { avgRating: { '$gt': 4 } },
    indexFilterSet: false,
    queryHash: '85BA8276',
    planCacheShapeHash: '85BA8276',
    planCacheKey: '6EB4A3FD',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      inputStage: {
        stage: 'IXSCAN',
        keyPattern: { avgRating: -1 },
        indexName: 'avgRating_-1',
        isMultiKey: false,
        multiKeyPaths: { avgRating: [ ] },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
```

```
        direction: 'forward',
        indexBounds: { avgRating: [ '[inf.0, 4)' ] }
      },
      rejectedPlans: [ ]
    },
    executionStats: {
      executionSuccess: true,
      nReturned: 3,
      executionTimeMillis: 1,
      totalKeysExamined: 3,
      totalDocsExamined: 3,
      executionStages: {
        isCached: false,
        stage: 'FETCH',
        nReturned: 3,
        executionTimeMillisEstimate: 0,
        works: 4,
        advanced: 3,
        needTime: 0,
        needYield: 0,
        saveState: 0,
        restoreState: 0,
        isEOF: 1,
        docsExamined: 3,
        alreadyHasObj: 0,
        inputStage: {
          stage: 'IXSCAN',
          nReturned: 3,
          executionTimeMillisEstimate: 0,
```

```
},
serverInfo: {
  host: 'ac-5e4ljmv-shard-00-02.erhexq0.mongoddb.net',
  port: 27017,
  version: '8.0.29',
  gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
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  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1787022929, i: 1 }),
  signature: {
    hash: Binary.createFromBase64('2EaKjb2/ZDrylok0LT1DUQ0SY5Y=', 0),
    keyId: Long('7632047810002550786')
  }
},
operationTime: Timestamp({ t: 1787022929, i: 1 })
}
Atlas atlas-dklk63-shard-0 [primary] bookflow_db> █
```